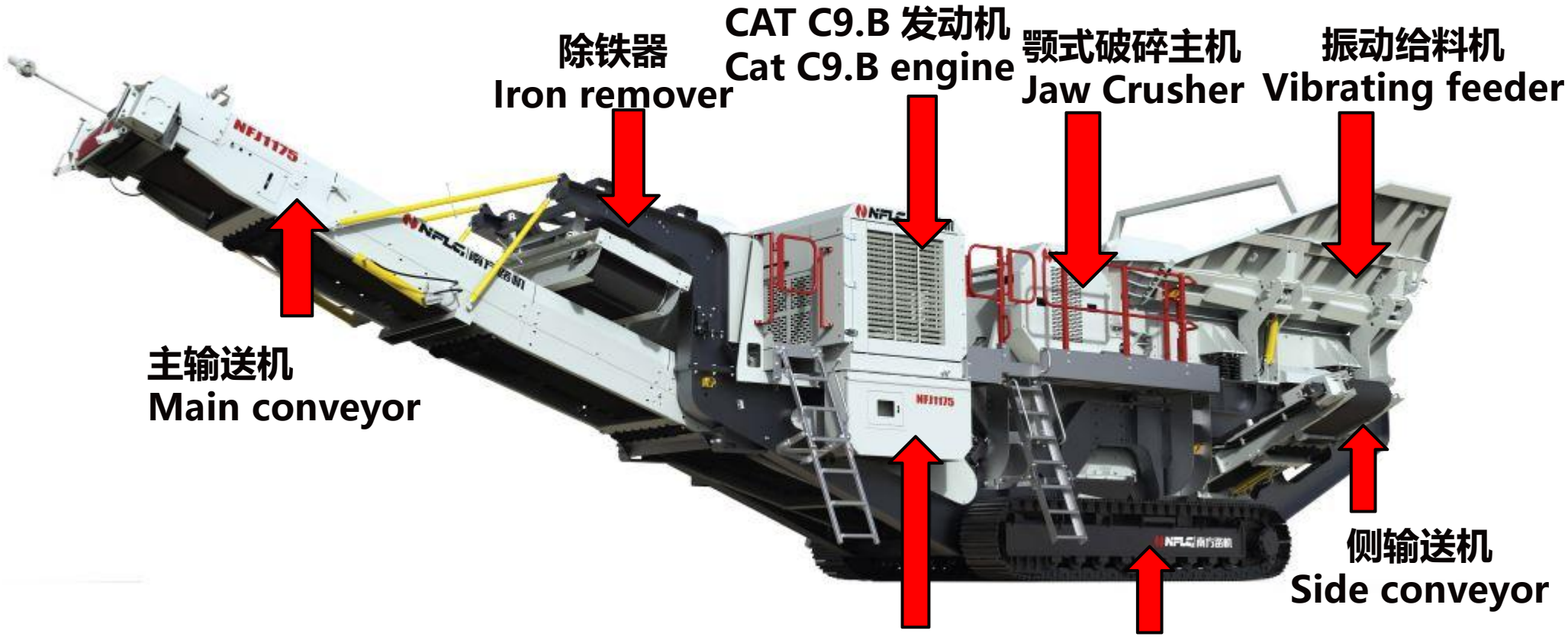
南方路机搅拌学院
NFLG MIXING INSTITUTE

移动破碎筛分事业部

NFJ1175设备维护资料
NFJ1175 equipment maintenance



1 整机部件介绍 Introduction to the whole machine parts



2 保养简介 Introduction to maintenance

	速度/Speed	流量/Flow	带宽/Bandwidth
主输送机/Main conveyor	110 米/分钟(m/min)	85 升/分钟	1200mm
取铁器/Iron remover	66 米/分钟(m/min)	54 升/分钟	750mm
侧皮带机/Side conveyor	57 米/分钟(m/min)	54 升/分钟	650mm
给料机/Feeder	450 – 850 转/分（Rpm）	80 升/分钟	X



整机重量（标准）
~~Machine weight~~ (standard) 54 吨/ton



柴油箱容积
Diesel tank volume 430L

液压油箱容积
Hydraulic oil tank volume 720 L



液压油:	ISO 46	0°C	+25°C
Hydraulic oil:	ISO 68	+20°	+40°C

履带:	VG 100	-20°C	+ 6°C
Track:	VG 150	+6°C	+30°C
	VG 320	+30°C	+50°C
给料机:	ISO 220	+4°C	+35°C
Feeder:	ISO 150	-18°C	+18°C

主机开口尺寸/ Crusher opening size : 760mm x 1070mm

最大进料尺寸/ Maximum feed size： 650mm



主机速度: 最小 230 转/分钟, 最大 280转/分钟
Crusher speed: minimum 230 rpm, maximum 280 rpm

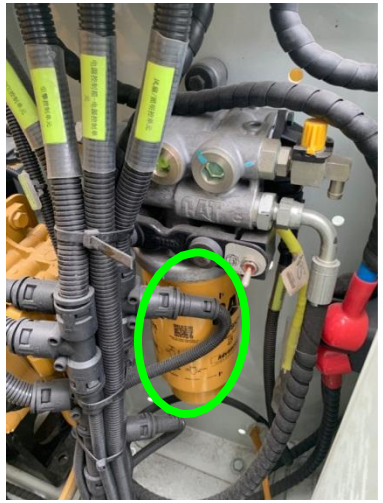
2 保养简介 Introduction to maintenance



润滑位置（Location）	添加量（capacity）	周期(Addition)
主输送机驱动滚筒轴承 main conveyor drive drum bearings	4g/PC	16H
主输送机尾部滚筒轴承 main conveyor tail drum bearings	16g/PC	50H
侧输送机所有滚筒轴承 all side conveyor drum bearings	4g/PC	16H
颚破主机动颚所有轴承 all moving jaw bearings	24g/PC	8H
颚破主机皮带轮轴承 jaw crusher belt pulley bearing	8g/PC	50H
除铁器轴承 all bearings of magnetic belt conveyor	4g/PC	50H

保养周期 Maintenance period		
发动机 (国三) Engine (National III)	500小时/H	机油滤芯 Oil filter
	500小时/H	燃油滤芯 Fuel filter
	500小时/H	油水分离器滤芯 Oil-water separator filter element
	500小时/H	空滤（外滤芯） Air filter (outer filter element)
	500 小时/H	空滤（内滤芯） Air filter (inner filter element)
液压系统 Hydraulic system	500 小时/H	回油过滤器 Return oil filter
	1000 小时/H	吸油过滤器(酌情更换) Suction filter (replace as appropriate)
发动机机油 Engine oil	500 小时/H	31.5 L(以实际液位为准 /Subject to actual liquid level)
冷却液 Engine coolant	3000 小时/H	40.9 L(以实际液位为准 /Subject to actual liquid level)
履带齿轮箱 Track gear box	1000 小时/H	10 L(以实际液位为准 /Subject to actual liquid level)
给料机齿轮箱 Feeder gear box	500 小时/H	8 L(以实际液位为准/Subject to actual liquid level)

3 动力系统 Power System



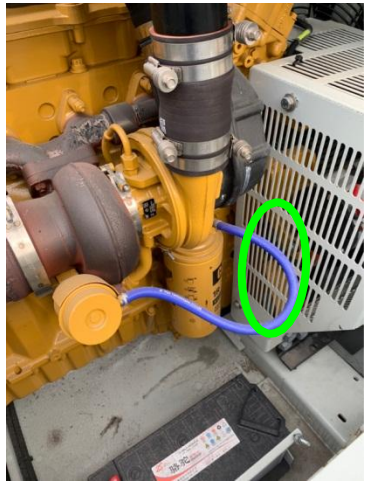
油水分离器滤芯
Oil-water separator filter element



燃油初级过滤器滤芯
Primary fuel filter element



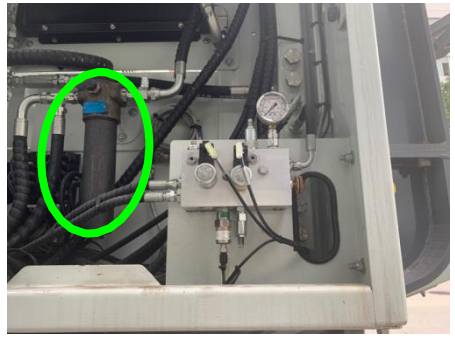
燃油二级过滤器滤芯
Secondary fuel filter element



机油滤芯
Oil filter



空气过滤器滤芯
Air filter element



高压过滤器滤芯
High pressure filter element



吸油过滤器滤芯
Oil suction filter element

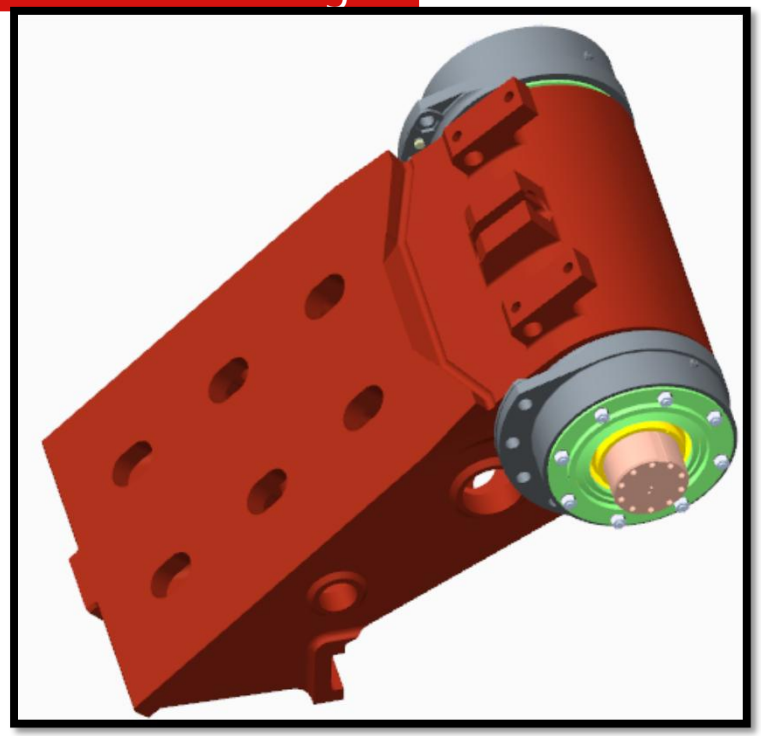


回油过滤器滤芯
Return oil filter element



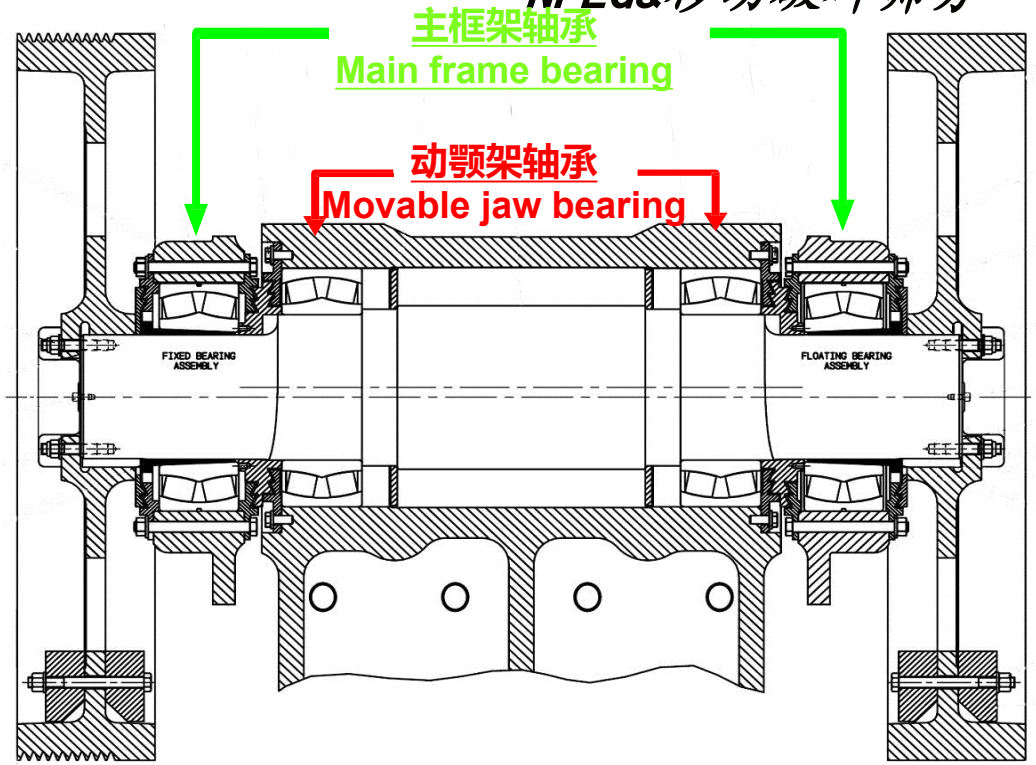
油箱呼吸过滤器
Fuel tank breathing filter

4 颚破主机轴承 Jaw crusher bearing



- 主机上一共四个轴承需要润滑，如上图
- 润滑要求：每个轴承8个小时加注24克，使用壳牌、福斯EP2润滑脂或者同等标准的其他品牌
- 需要边运转主机边打黄油，这样可以确保黄油可以润滑整周轴承
- 每隔500小时需要黄油打到溢出迷宫环，这样可以很好的起到密封作用，防止轴承损坏。颚破主机轴承的密封形式为迷宫式密封，也叫非接触式密封。迷宫内部间隙靠黄油填充起到密封作用。

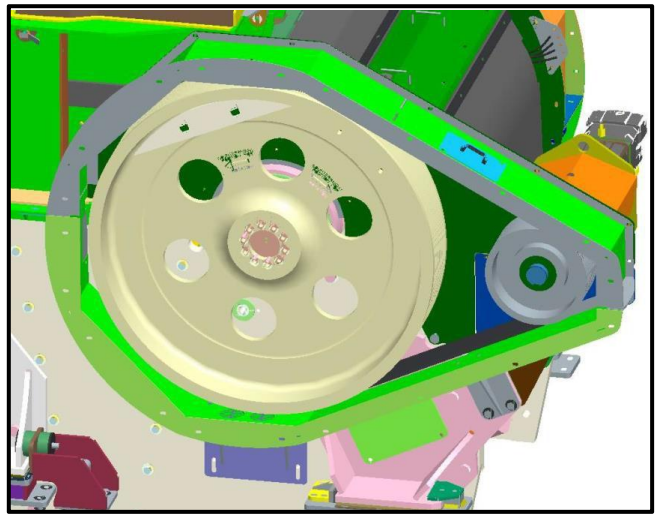
NFLG&移动破碎筛分

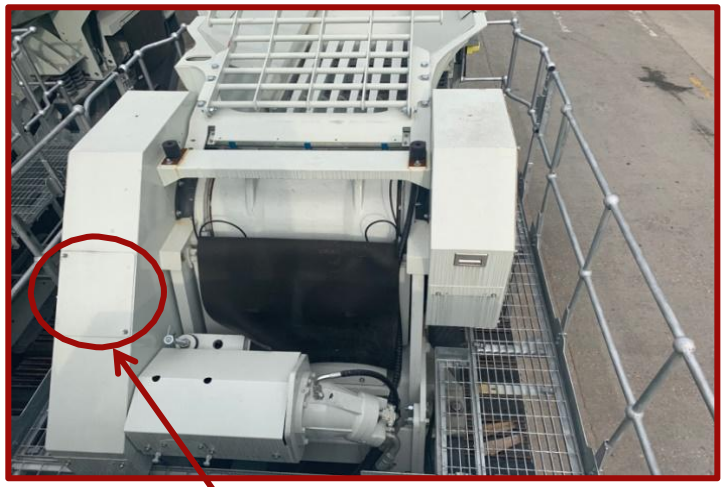


- A total of four bearings on the host need to be lubricated, as shown above
- Lubrication requirements: each bearing is filled with 24 grams in 8 hours, using Shell, Flowserve EP2 grease or other brands of equivalent standards.
- It is necessary to apply butter while running the main engine, so as to ensure that the butter can lubricate the bearings throughout the entire circumference.
- Every 500 hours, you need to add butter to the labyrinth ring to the state where it is overflowing., which can play a good sealing role and prevent bearing damage. The sealing form of the bearing of the jaw crusher is a labyrinth seal, also called a non-contact seal. The gap inside the labyrinth is sealed by butter filling.

5 主机驱动 Crusher driver

02-01 皮带张紧/The tension of belts





观察口
Observation port

注意：

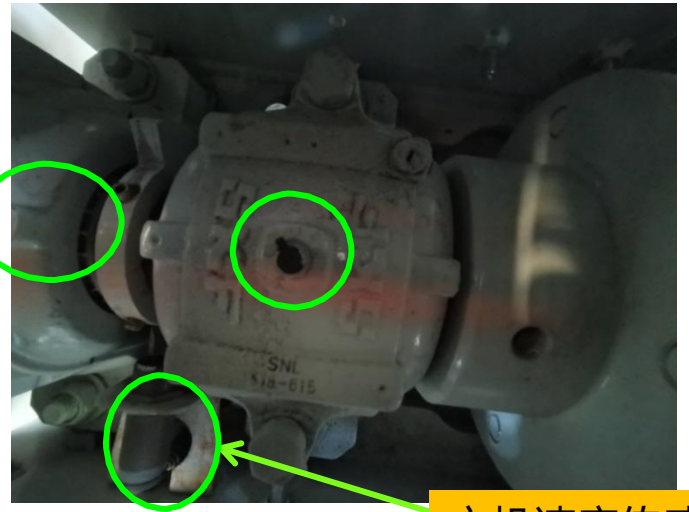
新设备或者新安装的皮带每个2、4、8小时需要检查皮带的张进度，以免运行过程导致皮带损坏。

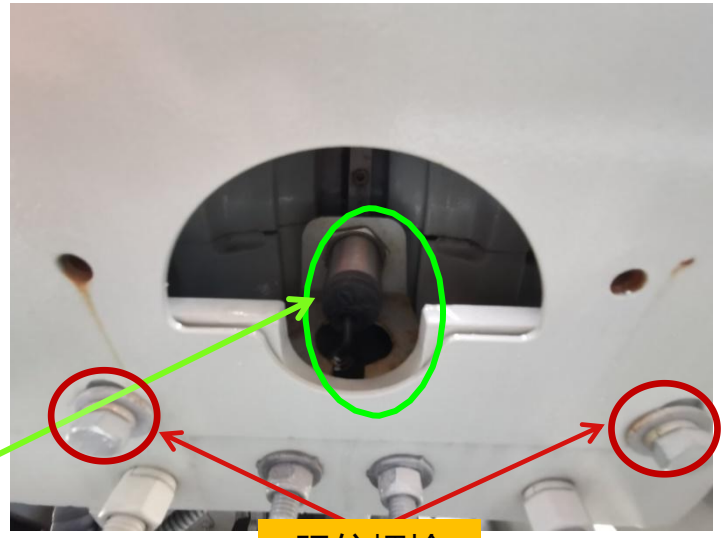
Notice:

New equipment or newly installed belts need to be checked the tension of the belts every 2, 4, 8 hours to avoid damage to the belts during operation.

5 主机驱动 Crusher driver

02-02 润滑、调整/ Lubrication and adjustment



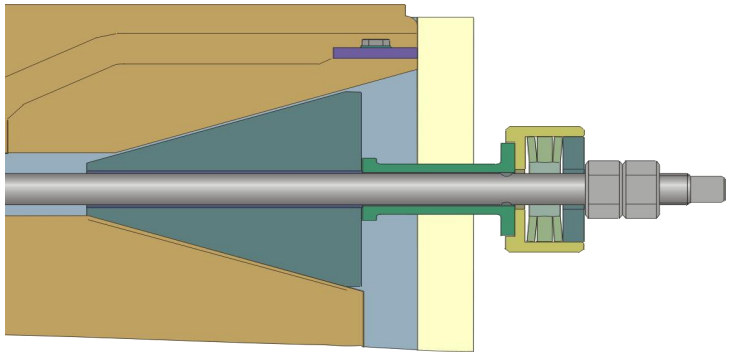


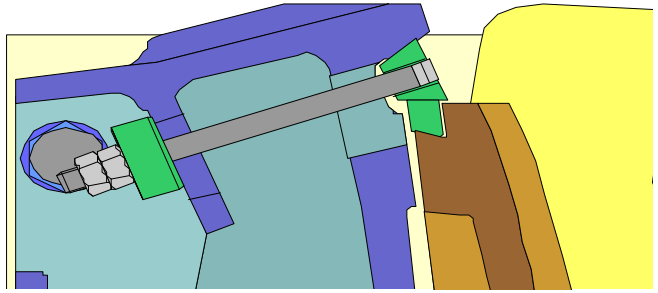


- 每个轴承的润滑为每50小时4克，不可过量打油会导致轴承温度过高导致轴承损坏。黄油使用壳牌或者美孚EP2标准或者同等级别的其他品牌
- 主机速度传感器上要保持清洁，否则无法检测到信号会导致“主机速度低”报警
- 主机速度传感器与凸齿的距离在8mm以内否则检测不到信号，调整距离时注意防止“打齿”
- 定时检查限位螺栓的紧固情况，一旦发生松动，需要重新调整马达与轴承的同轴度，具体调整方法咨询售后部门

- The lubrication of each bearing is 4 grams per 50 hours. Excessive lubrication should cause the bearing temperature to be too high and cause the bearing to be damaged. Butter uses Shell or Mobil EP2 standard or other brands of equivalent level
- The speed sensor of the crusher must be kept clean, otherwise the failure to detect the signal will cause the "host speed low" alarm
- The distance between the crusher speed sensor and the convex tooth is within 8mm, otherwise the signal will not be detected. When adjusting the distance, pay attention to prevent "tooth hitting"
- Regularly check the tightening of the limit bolts. Once loosened, the coaxiality of the motor and the bearing needs to be re-adjusted. Consult the after-sales department for the specific adjustment method.

6 主机衬板 Crusher liner





动颚张紧杆
Movable jaw tension rod



定颚张紧杆
Fixed jaw tension rod

主机衬板在磨损至极限后需调头或者更换
重新装配衬板后，每工作2，4，8小时加紧一次衬板张紧装置
The crusher liner needs to be turned or replaced after it is worn to the limit.
After re-assembling the liner, tighten the liner tensioning device every 2, 4, 8 hours of work

7 液压风扇 Hydraulic fan



- 由于现场粉尘大，需要定时对风扇进行清洗，防止出现液压油温度高
- 风扇采用的是吹风形式，重新安装时需要确认风向
- 机器工作时候，需要检查风扇是否正常运转

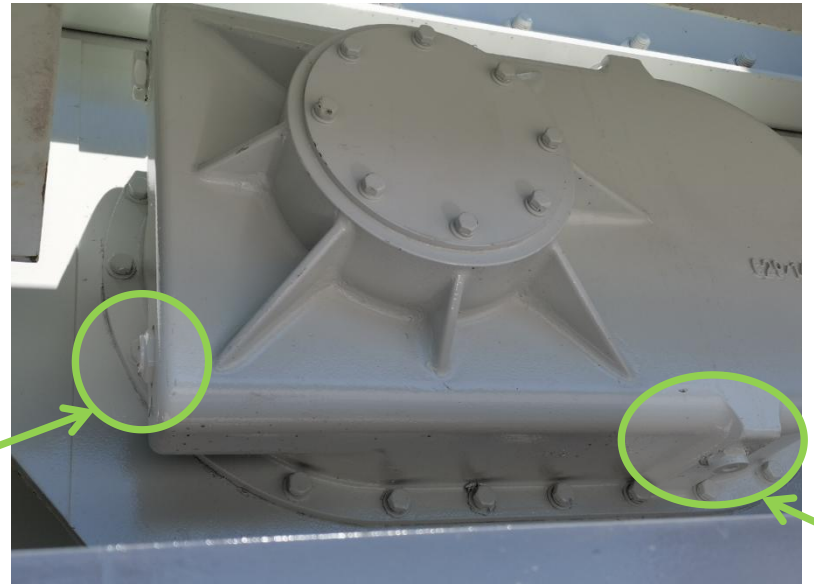
- Due to the large amount of dust on site, the fans need to be cleaned regularly to prevent high hydraulic oil temperatures
- The fan adopts the blowing mode, and the wind direction needs to be confirmed when reinstalling
- When the machine is working, you need to check whether the fan is running normally

8 给料机 Feeder

08-01 润滑/Lubrication



油温指示，保证油液在油镜中间位置以下
Oil temperature indicator to ensure that the oil is below the middle position of the oil glass





放油口位置
Oil drain position

- 振动单元的润滑油必须达到ISO等级：220 (5°C to 40°C)，150 (-15°C to 5°C)，320 (40°C 以上)，由你处所具体环境温度而定
- 开始250小时更换，后面每隔500小时更换，驱动侧的加油量3.8升，非驱动侧加油量也为3.8升

- The lubricating oil of the vibration unit must reach ISO grade: 220 (5oC to 40oC), 150 (-15oC to 5oC), 320 (above 40oC), depending on the specific ambient temperature of your location
- Replace after 250 hours at the beginning, and every 500 hours later. The refueling volume on the drive side is 3.8 liters, and the refueling volume on the non-drive side is also 3.8 liters.

8 给料机 Feeder

08-02 更换齿轮油/Change gear oil



正常工作时油液位须添加达到油标中心红点。
During normal operation, the oil level must be added to reach the red point of the oil mark center.



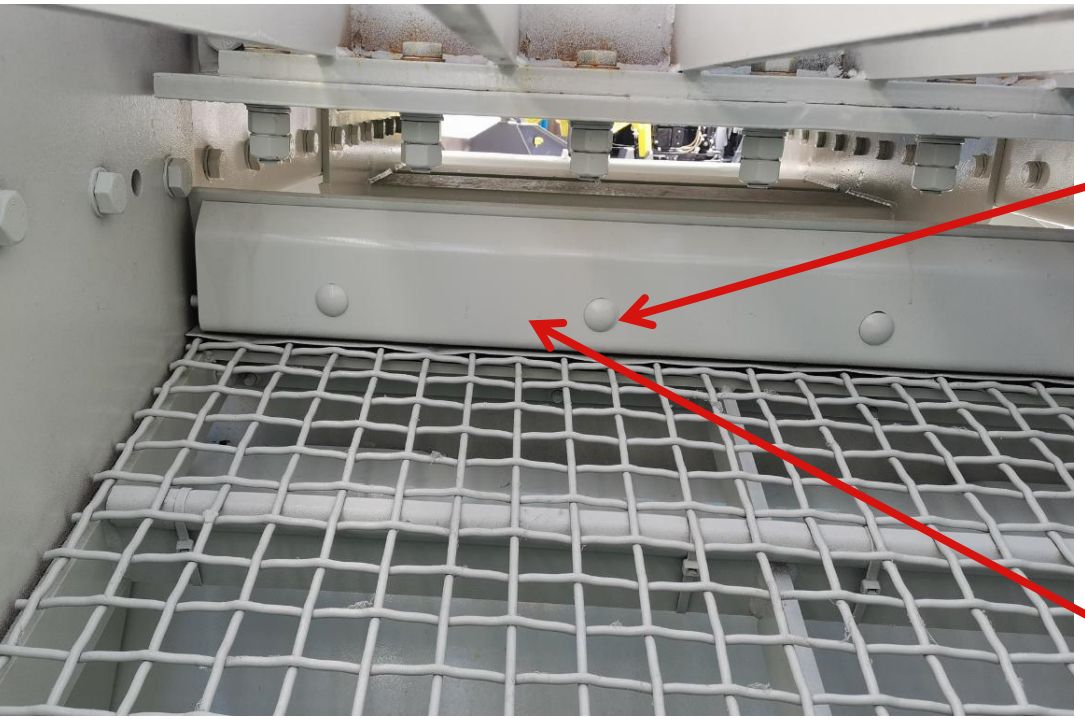


排油口
oil drain port

更换齿轮油时需要打开放油口跟排油口把密闭空间内的旧油放干净.
When changing gear oil, you need to open the oil drain port to drain the old oil in the enclosed space.

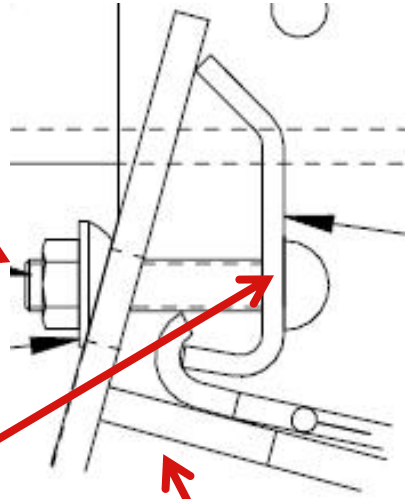
8 给料机 Feeder

08-03 筛网张紧/Screen tensioning

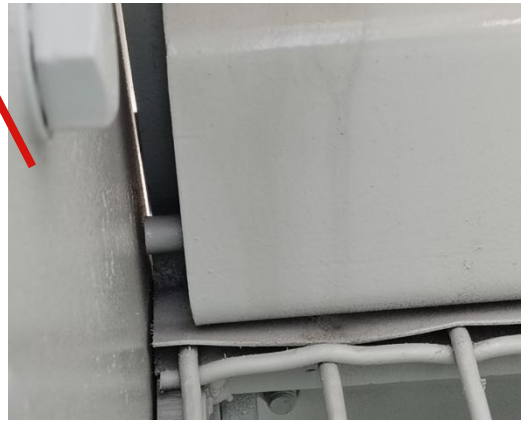




张紧螺栓
Tension bolts



张紧板
Tension plate

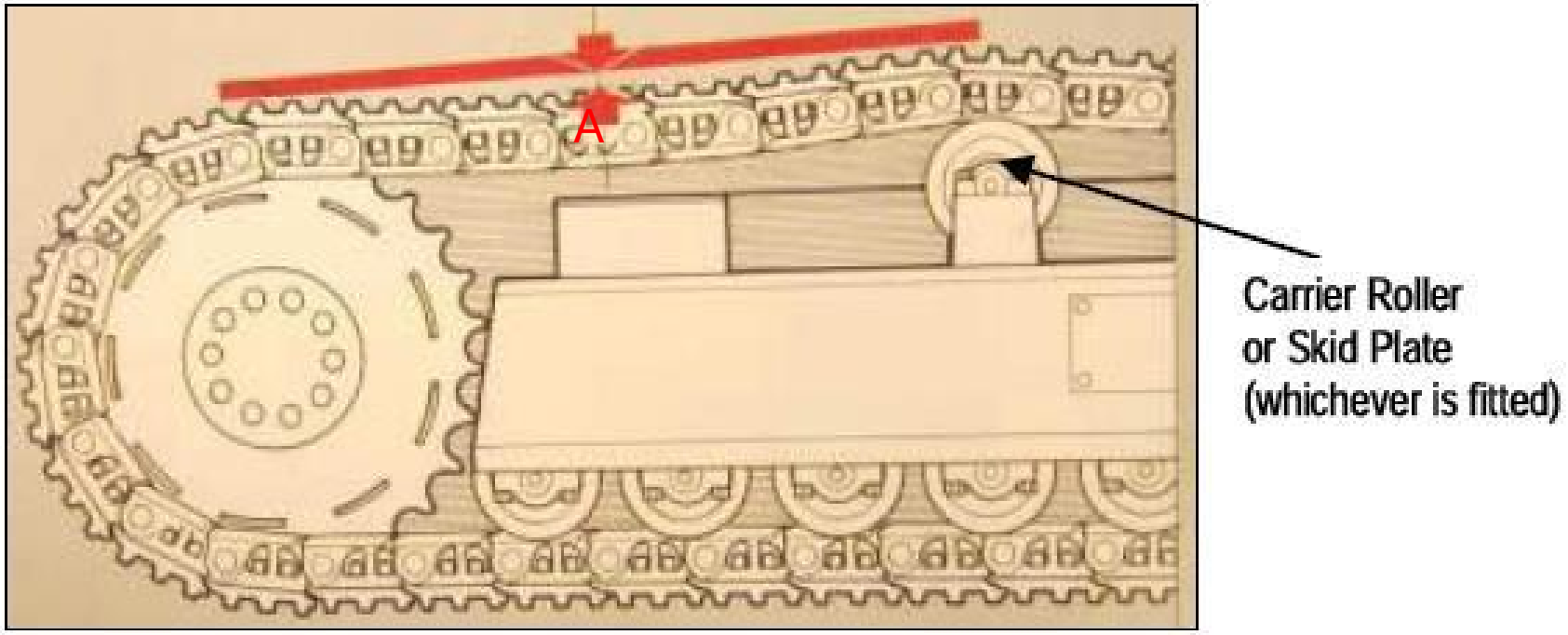


注意：当新更换筛网时，每隔2小时、4小时、6小时需要对筛网进行张紧。避免筛网未张紧导致筛网损坏。

Note: When a new screen is replaced, the screen needs to be tensioned every 2 hours, 4 hours, and 6 hours. Avoid damage to the screen due to untensioned screens.

9 履带行走 Track walking

09-01 履带张紧/Track tension



The sag of the track must be between 5mm and 15mm .

把履带开到硬实、水平的地面，向前开2米左右（最少），测量无支撑处链条的最大下垂距离，确保距离在5-15mm之间。

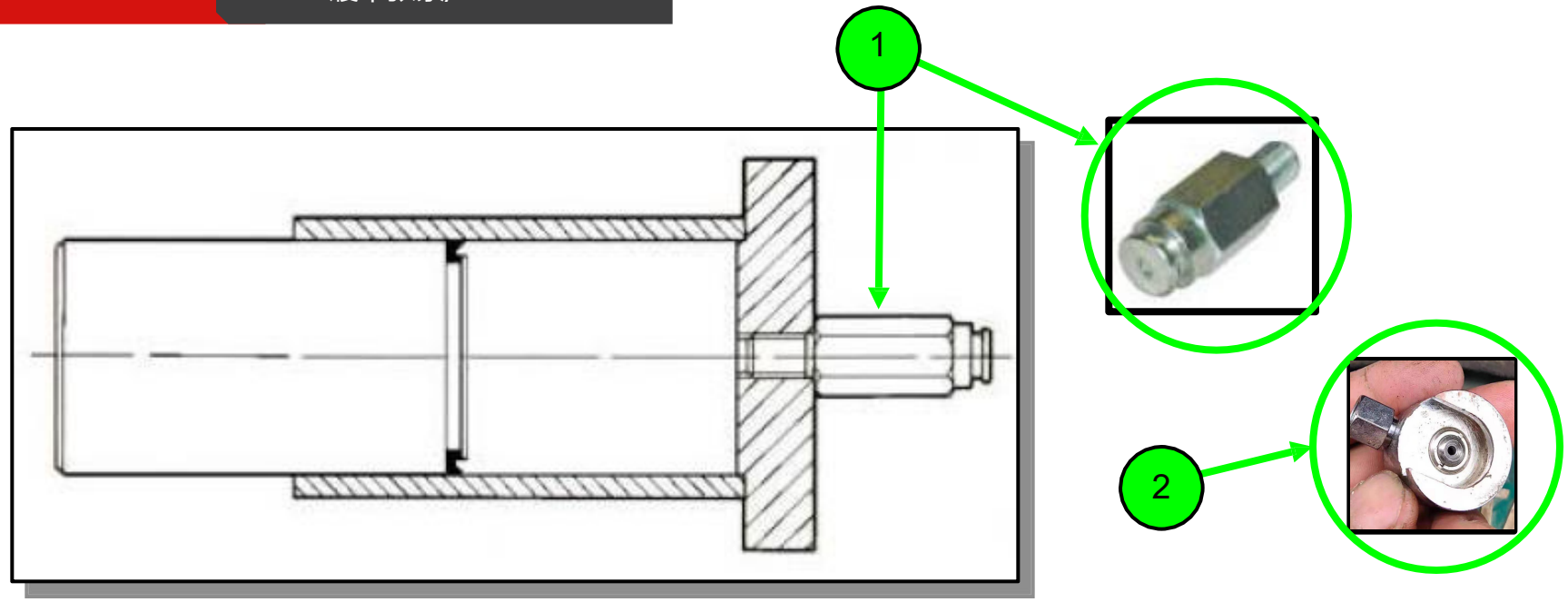
如上图所示

Drive the track to a hard, level ground, drive forward about 2 meters (at least), and measure the maximum sag distance of the chain where there is no support to ensure that the distance is between 5-15mm.

As shown in the picture above

9 履带行走 Track walking

09-01 履带张紧/Track tension



- 履带系统用一个油脂缸来保持每个履带的张紧力。黄油嘴旋紧于油脂缸尾部，如图1所示，通过给润油箱泵油和放油，可以分别拉紧和松弛履带。
- 如图2所示，油脂适配器是与黄油嘴配对使用，它存放在工具箱中。
- 注：履带张紧轮里的油脂是有压力的，在旋松黄油嘴的时候要特别小心。

- The track system uses a grease tank to maintain the tension of each crawler. The grease nipple is screwed to the end of the grease tank, as shown in Figure 1. By pumping and draining oil to the lubricating tank, the track can be tightened and loosened respectively.
- As shown in Figure 2, the grease adapter is paired with the grease nipple, and it is stored in the tool box.
- Note: The grease in the track tensioner is under pressure, so be especially careful when loosening the grease nipple.

9 履带行走 Track walking

09-01 履带张紧/Track tension

图1(Figure 1)



图2(Figure 2)



图3(Figure 3)

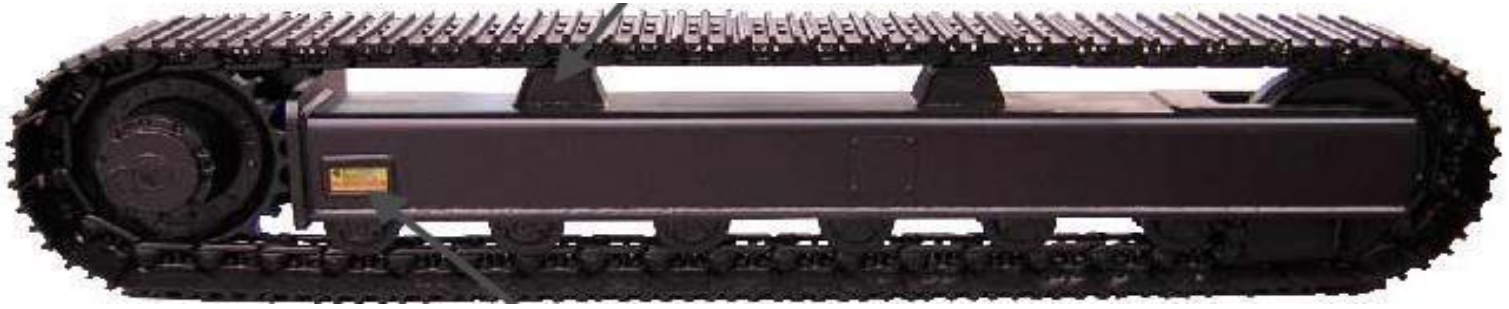


- 旋松2个螺钉，打开盖子如图1、图2所示
- 确保黄油嘴和黄油枪是干净的，有污垢进入黄油嘴会导致故障。如图3所示，把黄油枪连接到黄油嘴上，打进黄油一直达到履带张紧力所给定值。
- 机器向后行驶50米，如果履带还是松弛的话，重复上面的步骤。

- Loosen the 2 screws and open the cover as shown in Figure 1 and Figure 2.
- Make sure that the grease nipple and grease gun are clean. Dirt entering the grease nipple will cause malfunction. As shown in Figure 3, connect the grease gun to the grease nipple, and inject the grease until the specified value of the track tension is reached.
- The machine travels backward 50 meters, if the track is still slack, repeat the above steps.

9 履带行走 Track walking

09-02 润滑/Lubrication



LUBRICANT	-20°C + 5°c IV 95 min	+5°C + 30°c IV 95 min	30°C + 50°c IV 95 min	30°C + 65°c IV 95 min
ESSO	Spartan EP 100	Spartan EP 150	Spartan EP 320	Compressor oil LG 150
AGIP	Blasia 100	Blasia 150	Blasia 320	Blasia S 220
ARAL	Degol BG 100	Degol BG 150	Degol BG 320	Degol BS 220
BP MACH	GR XP 100	GR XP 150	GR XP 320	SGR XP 200
CASTROL	Alpha SP100	Alpha SP150	Alpha SP 320	Alpha SN6
ELF	Reduetell SP 100	Reduetell SP 150	Reduetell SP 320	Ortils 125 MS Syntherma P30
CHEVRON	Non leaded Gear compound 100	Non leaded Gear compound 150	Non leaded Gear compound 320	
GULF		EP lubricant HD 150	EP lubricant HD 320	
I.P	Mellana 100	Mellana 150	Mellana 320	Telesia oil 150
MOBIL		Mobilgear 629	Mobilgear 632	Glycoyle 22/30 SHC 630
SHELL	Omala oil 100	Omala oil 150	Omala oil 320	Tivela oil SA
TOTAL	Carter EP 100N	Carter EP 150	Carter EP 320N	
KLUBER	Lamora 100	Lamora 150	Lamora 320	
ISO 3448	VG 100	VG 150	VG 320	VG 150-200

开始的第100个小时更换齿轮油，后面每隔1000小时更换，容量为每边10升。

Change the gear oil in the first 100 hours, and then change it every 1000 hours, with a capacity of 10 liters per side.

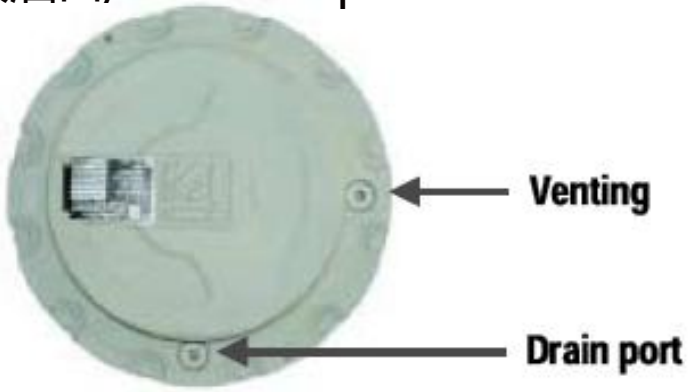
注：齿轮油的牌号及其品牌根据左边提供的进行购买

Note: The gear oil grades and brands should be purchased according to the one provided on the left

9 履带行走 Track walking

09-02 润滑/Lubrication

放油口/Oil drain port



如图所示，排放油时，需使机器中的一个堵塞处于6点钟位置，旋开两个堵塞使所有的润滑油排放到一个合适的容器中。最后安全合理地处理掉这些排污油。

As shown in the figure, when draining the oil, one of the blockages in the machine should be at the 6 o'clock position, and the two blockages should be unscrewed to discharge all the lubricating oil into a suitable container. Finally, dispose of these sewage oil safely and reasonably.

加油口/Oil fill port

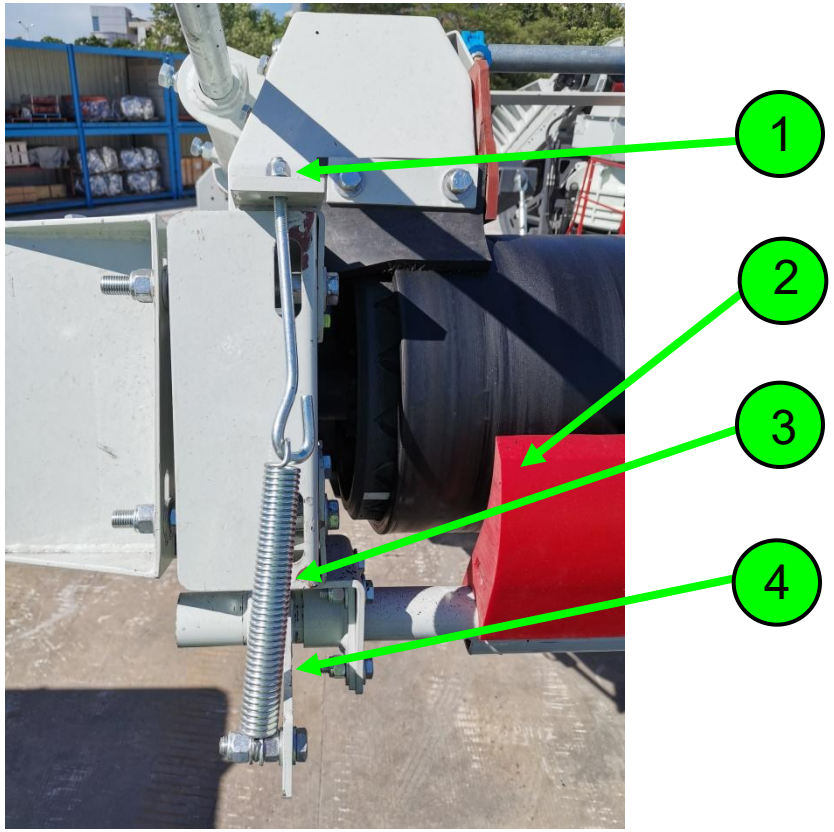


如图所示：填加润滑油时，需使机器的变速箱处于水平位置，使其中一个堵塞处于12点钟方向。旋开两个堵塞，从最上面的塞孔加油，直到油位达到最低的那个塞孔位。

As shown in the figure: when filling in lubricating oil, the gearbox of the machine should be in a horizontal position, and one of the blockages should be in the 12 o'clock position. Unscrew the two plugs and add oil from the top plug hole until the oil level reaches the lowest plug hole position.

10 主输送机 Main conveyor

10-01 头部刮料器 /Head scraper



1.弹簧调整螺母/Spring adjustment nut

2.聚氨酯刮板/Polyurethane scraper

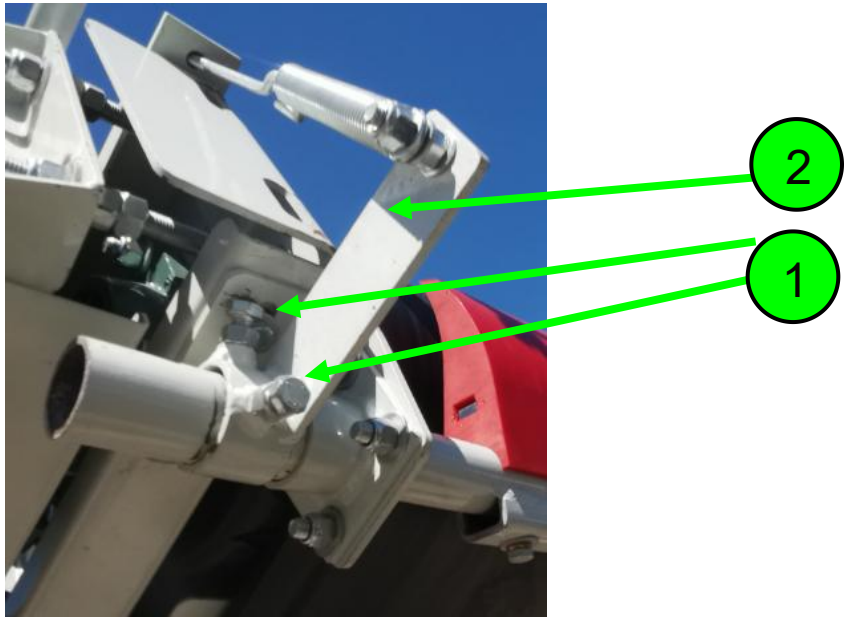
3.拉伸弹簧/Tension spring

4.调整臂/Adjusting arm

- 如果弹簧的张紧力变松的话，当你把它向上调整时，调整臂螺栓会碰到输送机骨架，因此为了达到正常的张紧力，你必须对调整臂进行在调整。
- If the tension of the spring becomes loose, when you adjust it upward, the adjusting arm bolts will touch the conveyor frame, so in order to achieve the normal tension, you must readjust the adjusting arm.
- 当调整刮料器正常拉紧力时，要确保刮料器的背面是干净的，然后锁紧1#调整螺母直到加载在弹簧上的力刚好靠你单手可以将调整臂翻转向下。
- When adjusting the normal tension of the scraper, make sure that the back of the scraper is clean, and then lock the 1# adjusting nut until the force loaded on the spring is just up to you to turn the adjusting arm down with one hand.

10 主输送机 Main conveyor

10-02 头部刮料器调整/ Adjust head scraper



1.调整臂固定螺栓/Adjusting arm fixing bolts

2.调整臂 / Adjusting arm

- 如果弹簧的张紧力变松的话，当你把它向上调整时，调整臂螺栓会碰到输送机骨架，因此为了达到正常的张紧力，你必须对调整臂进行在调整。
- If the tension of the spring becomes loose, when you adjust it upward, the adjusting arm bolts will touch the conveyor frame, so in order to achieve the normal tension, you must readjust the adjusting arm.
- 首先你需要释放弹簧的张紧力，旋松2个调整臂固定螺栓，然后使刮料器稳固地靠着皮带，再稍微降低调整臂直到有足够的空间以备正常的调整。最后锁紧螺栓，重新调整弹簧张紧力。
- First you need to release the tension of the spring, loosen the 2 adjusting arm fixing bolts, and then make the scraper firmly against the belt, and then slightly lower the adjusting arm until there is enough space for normal adjustment. Finally, tighten the bolts and readjust the spring tension.

10 主输送机 Main conveyor

10-03 头部支撑调整/Head support adjustment

图1(Figure 1)



图2(Figure 2)

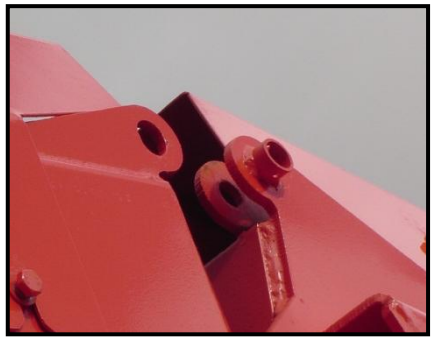


图3(Figure 3)



图4(Figure 4)



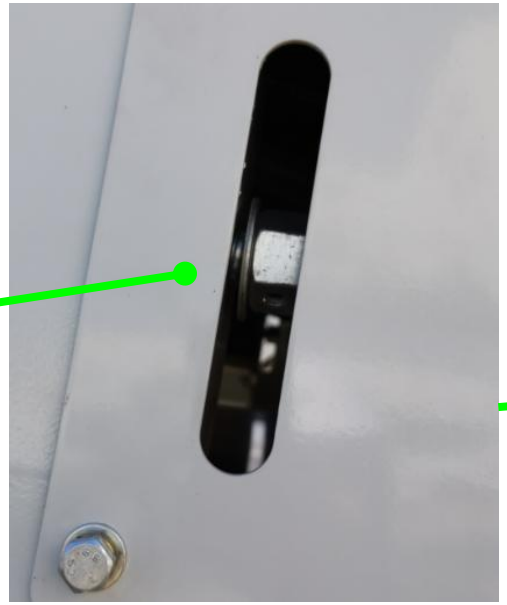
- 图1：确保左右两边的拉紧器拉紧皮带机，同时保证皮带机左右水平，不然会造成主皮带头部晃动厉害。
- 图2、图3：保证销轴固定到位。
- 图4：确保支撑横梁上面的6个螺栓没有发生松动。

- Figure 1: Ensure that the tensioners on the left and right sides tighten the belt conveyor, and at the same time ensure that the belt conveyor is horizontal, otherwise it will cause the main belt head to shake severely.
- Figure 2 and Figure 3: Ensure that the pin is fixed in place.
- Figure 4: Make sure that the 6 bolts on the supporting beam are not loose.

10 主输送机
Main conveyor

10-04 头部滚筒调整/Head roller adjustment







通过此位置对皮带头部滚筒进行调整
Adjust the belt head roller through this position

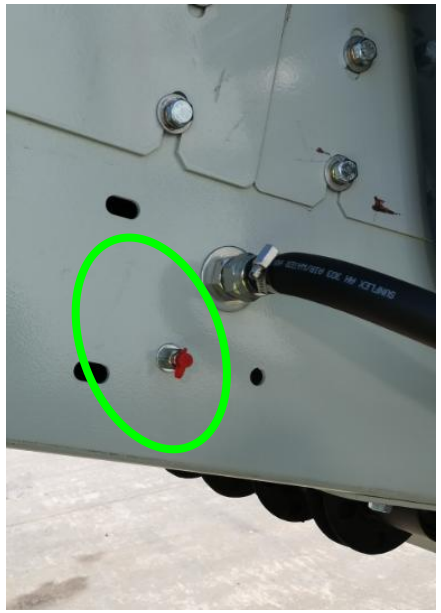
10 主输送机 Main conveyor

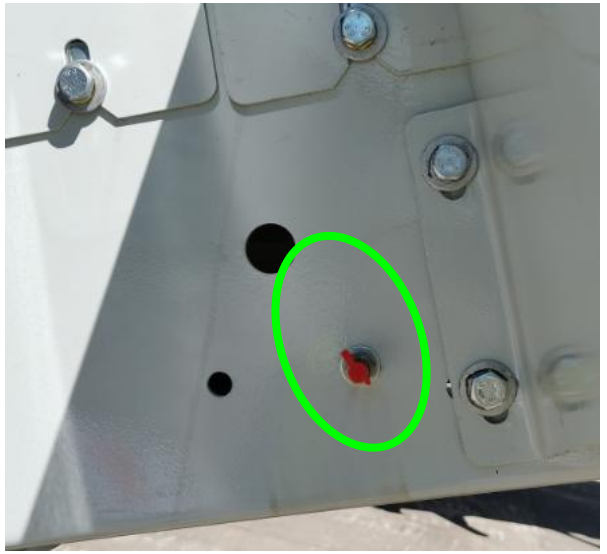
10-05 滚筒轴承润滑/Roller bearing lubrication



主皮带机轴承低位润滑点

Low lubrication point of main belt conveyor bearing





主输送机前后滚筒轴承每工作16小时需润滑4g

The front and rear roller bearings of the main conveyor need to be lubricated 4g every 16 hours of work

10 主输送机 Main conveyor

10-05 尾部档边橡胶调整、更换/Adjustment and replacement of the rear rubber plates





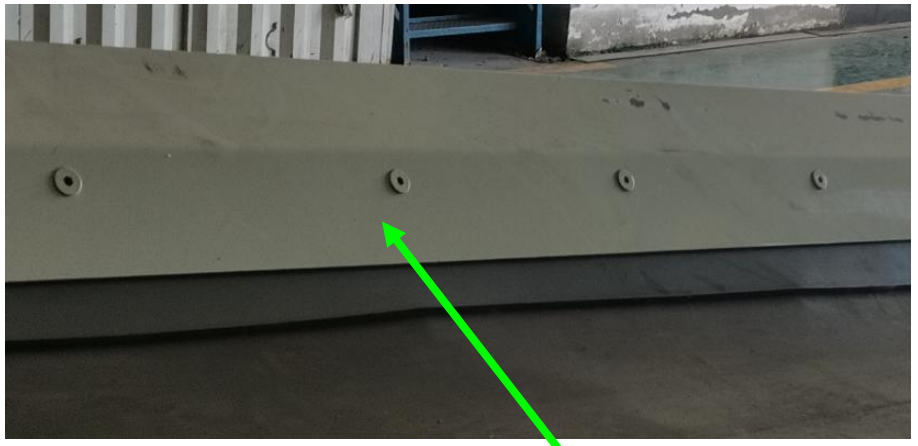




图1、图2：皮带机运行过程中需对尾部的档边橡胶进行的及时调整。如果档边橡胶没有安装到位，石料进入皮带内部，可能导致皮带损坏。

Figure 1 and Figure 2: Timely adjustment of the rubber at the rear of the belt conveyor is required during the operation of the belt conveyor. If the rubber is not installed in place, the stones will enter the belt, which may cause damage to the belt.

10 主输送机 Main conveyor

10-06 侧档边橡胶的调整、更换 /Adjustment and replacement of side rubber

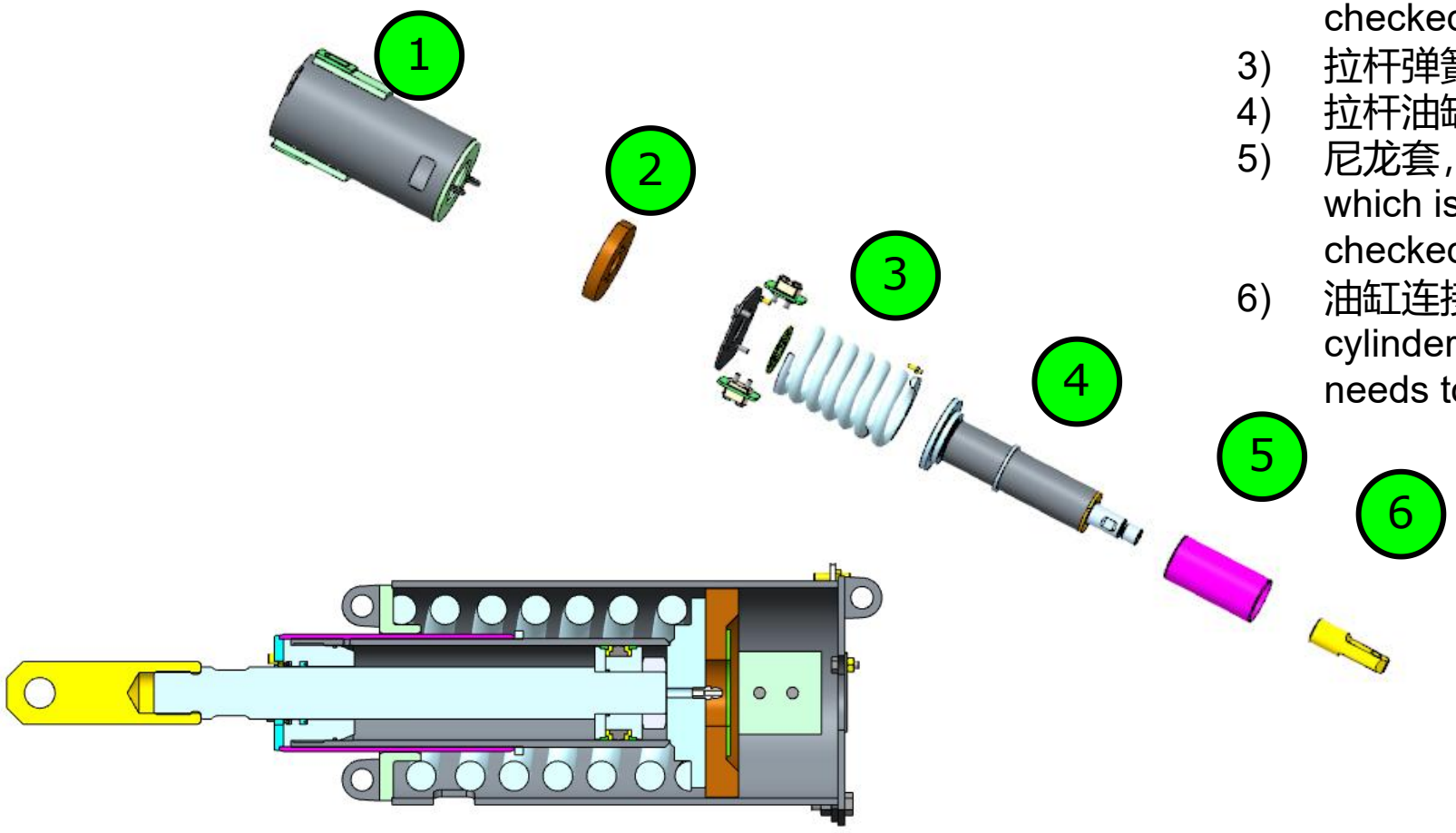


一段时间，档边橡胶发生磨损后就需要进行调整或者更换。松开皮带架子的螺栓就可以对橡胶进行调整。

After a period of time, the side rubber needs to be adjusted or replaced after it has been worn out. The rubber can be adjusted by loosening the bolts of the belt frame.

11 主机液压拉杆
Crusher hydraulic pull rod

组装图/Assembly drawing



1) 拉杆壳体/Pull rod shell
2) 尼龙块，属于易损件需定期检查更换/Nylon block, which is a vulnerable part and needs to be checked and replaced regularly
3) 拉杆弹簧/Pull rod spring
4) 拉杆油缸/Pull rod oil cylinder
5) 尼龙套，属于易损件需定期检查更换/Nylon sleeve, which is a vulnerable part and needs to be checked and replaced regularly
6) 油缸连接头，属于易损件需定期检查更换/ Oil cylinder connector, which is a vulnerable part and needs to be regularly inspected and replaced

12 液压油箱 Hydraulic oil tank

12-01 液压油液位/Hydraulic oil level

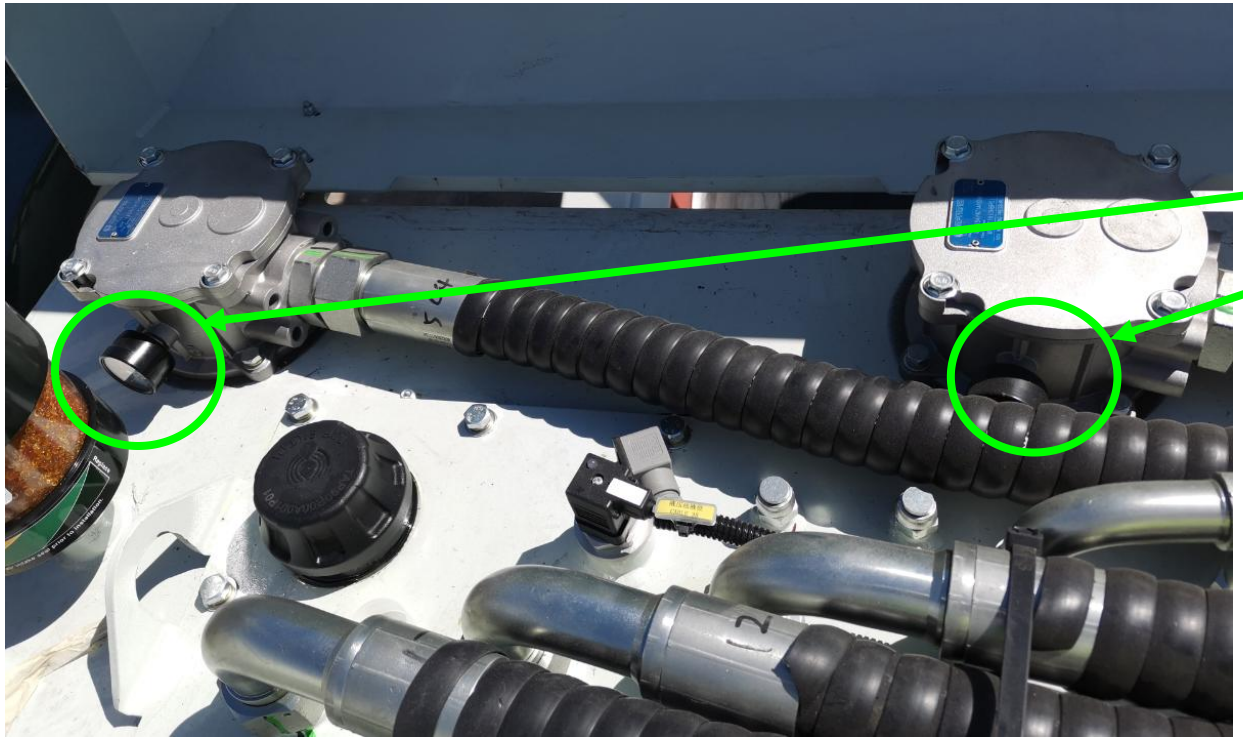


- 液压油箱的容积约为730升，带有吸油过滤器、回油过滤器(带堵塞指示器)
- 液压油箱油位建议不低于游标的中间刻度线，不得低于红色刻度线
- 油温不得超过85度，一旦超过就需要立即停机。正常的温度在60-70度之间
- 从油标中可以观察到油品的状况，一旦发生颜色变化就需要进行更换

- The volume of the hydraulic oil tank is about 730 liters, with suction filter and return filter (with clogging indicator)
- It is recommended that the oil level of the hydraulic oil tank should not be lower than the middle scale line of the vernier, and should not be lower than the red scale line
- The oil temperature should not exceed 85 degrees. Once it exceeds, the machine needs to be shut down immediately. The normal temperature is between 60-70 degrees
- The condition of the oil can be observed from the oil marker. Once the color changes, it needs to be replaced

12 液压油箱 Hydraulic oil tank

12-02 回油过滤器/Return oil filter





- 回油过滤器的过滤精度为10微米
- 如果发现指示器的指针指在红色的区间就需要进行进行更换过滤器
- 当已经使用超过500小时，即使只是堵塞指示器的指针还在绿色的区间也需要进行过滤器的更换

- The filtration accuracy of the return oil filter is 10 microns
- If you find that the pointer of the indicator is in the red zone, you need to replace the filter.
- When it has been used for more than 500 hours, even if the pointer of the clogging indicator is still in the green area, the filter needs to be replaced.

12 液压油箱 Hydraulic oil tank

12-03 空气呼吸器、油位开关/Air respirator, oil level switch



图 1 (Figure 1)

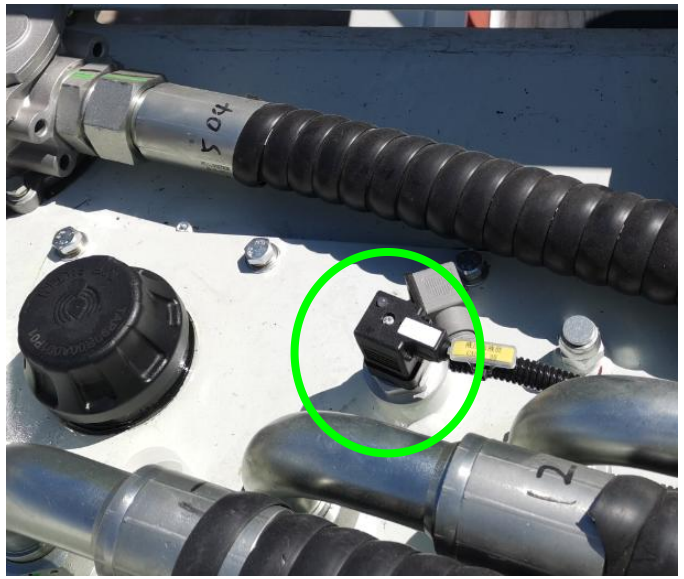


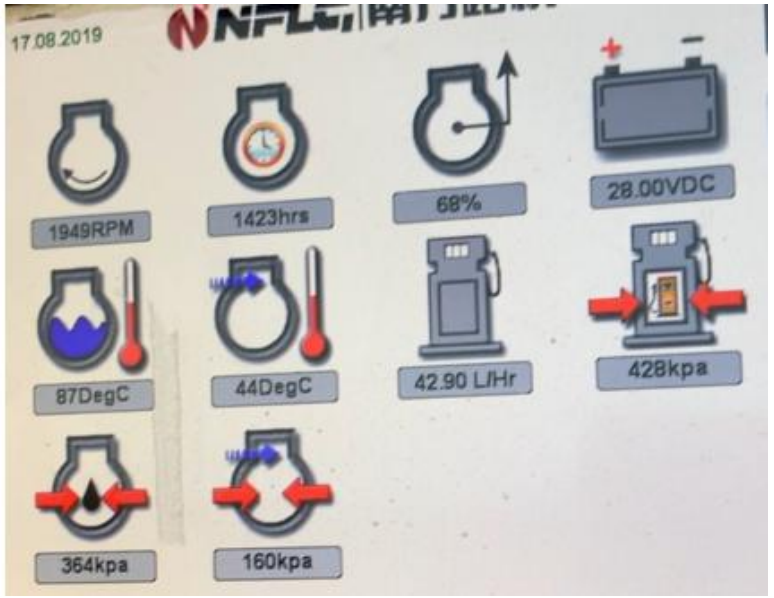
图 2 (Figure 2)

图1：呼吸器采用的是三级的过滤系统，防止水汽或者固体颗粒进入油箱内部。

图2：液压油箱油温开关，一旦低于正常值，机器就会停止工作，显示“液压油油位低”报警，需要立即添加液压油才能正常工作。

- Figure 1: The respirator uses a three-stage filter system to prevent water vapor or solid particles from entering the fuel tank.
- Figure 2: The hydraulic oil tank oil temperature switch, once it is lower than the normal value, the machine will stop working and display the "hydraulic oil level low" alarm. Hydraulic oil need to be added immediately to work properly.

13 发动机界面参数
Engine interface parameters



正常的参数范围（供参考）

发动机水温：85-90℃，113 ℃报警，116 ℃停机

进气温度：45-49℃，90℃报警

燃油压力：400-500kpa，与负载相关，758kpa报警

机油压力：300多kpa，154kpa报警，104kap停机

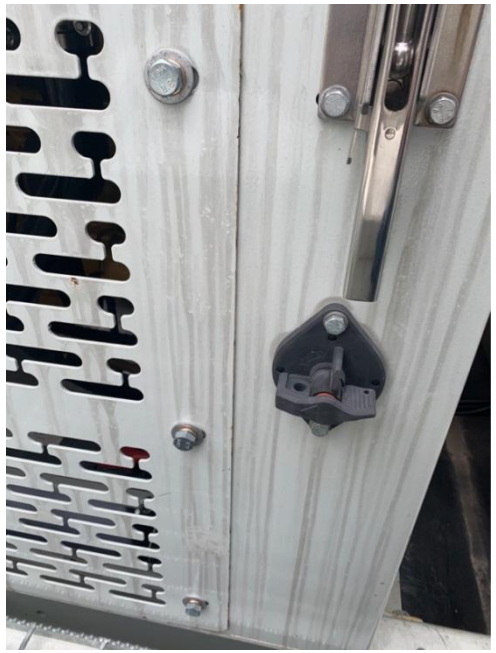
进气压力：正常工作时为160Kpa左右

Normal parameter range (for reference)

Engine water temperature: 85-90℃, alarm at 113℃, shutdown at 116℃
Inlet temperature: 45-49℃, 90℃ alarm
Fuel pressure: 400-500kpa, related to load, 758kpa alarm
Engine oil pressure: more than 300 kpa, 154 kpa alarm, 104 kpa shutdown
Inlet pressure: about 160Kpa during normal operation

14 保护操作
Protection operation

14-01 焊接操作/ Welding operation



关闭状态/Closed state

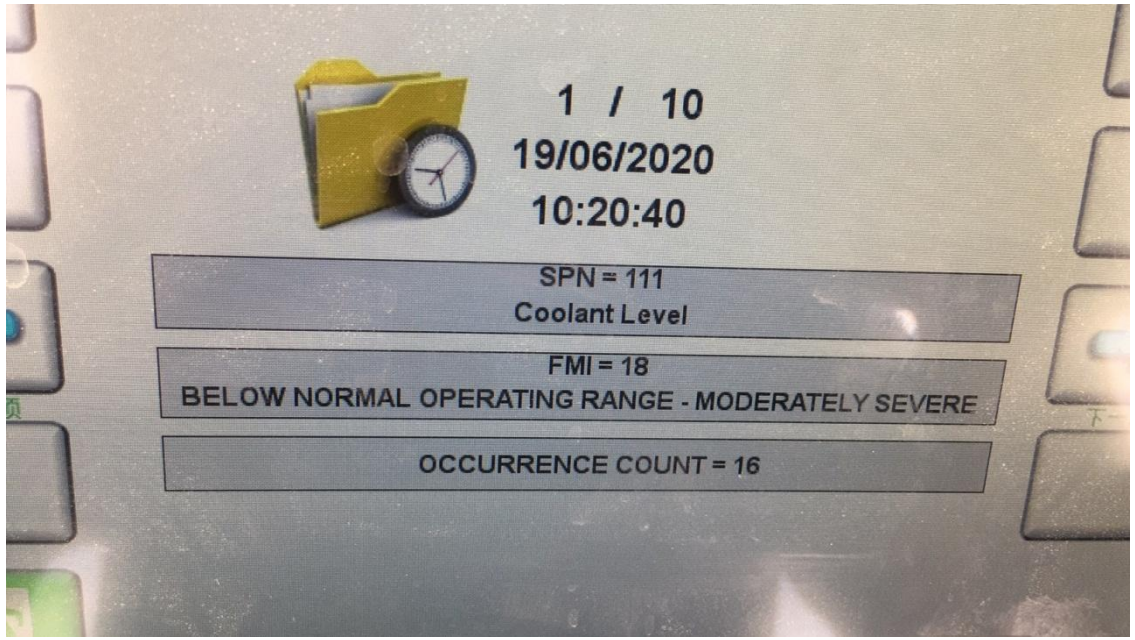


打开状态/Open state

在对设备进行焊接操作时，需要是隔离开关处于“关闭状态”，这点非常重要。
It is very important that the isolation switch is in the “closed state" when welding equipment.

15 故障处理程序
Fault handling procedures

16-01 发动机/Engine



冷却液位低可能原因：

Possible reasons for low coolant level:

- 插头松动/Loose plug
- 缺冷却液/Lack of coolant
- 水箱损坏，冷却液泄漏/The water tank is damaged and the coolant is leaking
- 传感器损坏/The sensor is damaged

第一步：当出现发动机故障报警时，需要提供发动机的故障代码（SPN）。然后结合故障描述判断是何种故障。

Step 1: When an engine failure alarm occurs, the engine's failure code (SPN) needs to be provided. Then combine the description of the fault to determine what kind of fault it is.

第二：确定导致此故障的可能原因

Step 2: Determine the possible cause of this failure

第三步：可能的原因按先易后难的处理顺序逐一排查

Step 3: Check the possible causes one by one in the order of easy first and then difficult

16 故障处理程序
Fault handling procedures

16-02 电气/Electrical system

故障代码	可能的原因	解决办法	故障图标&描述
1	•PLC 与显示屏通讯线接线错误或未连接。	✓检查接线，确认 CANH 与 CANL 正确连接。	 显示屏与 PLC 连接中断！
	•CANBUS 网络终端电阻未连接或安装错误。	✓断电状态下检查 CAN 网络上的电阻值，应为 60 欧姆。	
	•PLC 故障或程序未运行。	✓查看 PLC 状态灯，正常为 2Hz 绿色闪烁。	
2	•发动机 ECU 检测到自身存在某种故障。	✓查看产生发动机故障具体的 SPN 码和 FMI 码，以获取更加详细的故障信息，并及时与厂家联系。	 发动机故障！
3	•柜子侧边的 CANBUS 总线插头未连接或松动。	✓检查柜子侧边的 CAN 总线插头，确保连接正常。	 CANBUS 总线错误！
	•CANBUS 网络终端电阻未连接或安装错误。	✓断电状态下检查 CAN 网络上的电阻值，应为 60 欧姆。	
	•网络上的某个节点波特率设置不正确。	✓检查各节点的波特率设置，确保波特率一致。	
5	•设备急停按钮被按下动作。	✓检查设备上所有的急停按钮情况，右旋复位。	 设备急停故障！
	•急停按钮触点损坏或接触不良。	✓断电状态下，分别检查各个急停按钮触点的通断情况，必要时更换同型号急停按钮触点。	
	安全继电器损坏或触点接触不良。	✓更换同型号安全继电器。	
6	•有线遥控器插头或应急插头未连接。	✓检查柜子侧边的有线遥控器插头或应急插头，确保连接正常。	 有线遥控器急停故障！
	•有线遥控器急停按钮损坏。	✓更换同型号的急停按钮。	
	•有线遥控器急停按钮被按下。	✓检查遥控器上急停按钮情况，旋转复位。	

具体故障可参考
电气系统控制说明书

For specific faults, please refer to Electrical system control manual



感谢您的关注

Thank you for your attention

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